TIANYUE & TZBOT — AUTOMATIC CHARGING, BATTERY LIFE & COST

600 kg-class industrial AMR/AGV procurement research | Updated 30 September 2026

1. Executive summary

Both TIANYUE and TZBOT have published AGV/AMR solutions supporting automatic charging. For the user's 600 kg indoor material-handling target, automatic contact charging is the most straightforward option to investigate. The exact battery capacity and exact charging-station price for the specific 600 kg TIANYUE Mecanum AMR are not publicly disclosed, so this report clearly separates verified facts from estimates.

- TIANYUE 600 kg Mecanum AMR: public listing confirms LiFePO4 battery, 600 kg load, Mecanum drive, rear jacking, 3D camera and 60 m LiDAR, but does not publish battery Ah/kWh or operating hours for this exact model. [TIANYUE source]
- TIANYUE publishes comparable AGVs with 8 h endurance and automatic return-to-charge, and another 500 kg platform with 8–12 h per charge.
- TIANYUE's AGVC-48V150A-TY automatic charging pile is 48 V, up to 150 A, 380 V input, telescopic contact, infrared positioning and battery detection. Public price is quotation-only.
- TZBOT's 600 kg Laser-SLAM jacking AMR is specified with LiFePO4 battery and manual/automatic charging. Public pages do not state the battery Ah or operating hours for that exact 600 kg model.
- TZBOT also documents an automatic charging/scheduling system that manages docking and can maintain robots at about 70% battery.
- Public marketplace prices found earlier for TZBOT 600 kg AMRs were roughly US$11,800–18,600 depending on marketplace/configuration; these are robot prices, not necessarily the charger.

2. TIANYUE — automatic charging

The exact TIANYUE charging station requested is **AGVC-48V150A-TY**. The manufacturer specifies 48 V DC charging, up to 150 A, 380 V input, telescopic-contact charging, infrared positioning, battery checking, a 200 mm telescopic interface, wireless communication and an all-in-one design. MOQ is 1 and delivery is listed as 15 working days. The page states 3C/CE certification.

The charging station page does **not publish a fixed selling price**; it is quote-based.

TIANYUE charging station	Published specification
Model	AGVC-48V150A-TY
Charging voltage	48 V
Maximum charging current	150 A
Nominal maximum electrical output	48 × 150 = 7.2 kW (calculated)
Input	380 V AC
Charging interface	Telescopic contact
Telescopic interface	200 mm
Positioning	Infrared / photoelectric
Battery functions	Battery detection + charging protection
Communication	Wireless
MOQ / delivery	1 / 15 working days
Public price	Quote required

Source: https://www.roboticsagv.com/sale-44898002-integrated-48v-150a-agv-battery-charger-automatic-retractable-charging-pile.html

3. TIANYUE — battery life evidence

For the exact 600 kg Mecanum AMR, the public product page confirms a LiFePO4 battery but does not specify Ah capacity or endurance. Comparable TIANYUE products provide useful benchmarks: a 300 kg AGV lists 48 V/20 Ah and 8 h comprehensive endurance; another TIANYUE AGV lists 8 h battery life with automatic return-to-charge; a 500 kg platform lists 8–12 h per charge with a 48 V lithium battery.

TIANYUE reference	Battery / endurance	Charging
300 kg AGV	48 V / 20 Ah; 8 h	Manual / automatic / quick change
100–1000 kg AGV family	8 h battery life	Automatic return to charge
500 kg platform	48 V lithium; 8–12 h	Manual + automatic
1000 kg magnetic AGV	48 V / 40 Ah LiFePO4; 8 h	Automatic charging station
600 kg Mecanum target	LiFePO4 confirmed; Ah/h not published	Must confirm with factory

Important: “8 hours” is an operating/endurance specification, not necessarily 8 hours of continuous driving. Real endurance depends on load, speed, distance, lifting cycles, stops, floor condition and duty cycle.

4. TZBOT — automatic charging

TZBOT's 600 kg Laser-SLAM jacking AMR is specified with LiFePO4 battery and **manual/automatic charging**. Its safety system includes LiDAR, collision bar and emergency stop. The published 600 kg model uses differential drive, not Mecanum.

TZBOT also describes an AGV automatic charging/scheduling solution that automatically manages matching and docking of robots to charging piles and states that it can keep robots at about 70% power. This supports an opportunity-charging strategy for fleet operation.

5. TZBOT — battery life

For the exact 600 kg TZBOT SLAM jacking AMR, the public manufacturer page confirms LiFePO4 and automatic charging but does not publish Ah capacity or an operating-hours figure. A separate 2026 TZBOT laser-sensor AMR specifies 24 V/30 Ah and 6–8 h running time, demonstrating the type of battery/endurance data TZBOT publishes for some models, but that figure must **not** be applied to the 600 kg model.

TZBOT also has a 600 kg magnetic AGV with automatic/manual/battery-replacement charging, but it is a different model and uses magnetic navigation.

6. Cost research

Exact manufacturer prices for the TIANYUE 48 V/150 A charging pile and the batteries for the two target 600 kg AMRs were not publicly listed. The following are procurement estimates/benchmarks, not quotations.

Cost item	TIANYUE	TZBOT
600 kg AMR / robot	Exact price: quote required. Earlier comparable-market research	Public 600 kg market [illegible]
Automatic charging station	AGVC-48V150A-TY: quote required. Preliminary budget estimate	Exact [illegible]
Battery	Exact 600 kg battery Ah/kWh: quote required. LiFePO4 confirmed	Exact 600 kg battery Ah/kWh: quote required. LiFePO4 confirmed.
System package	Ask for AMR + battery + auto charger + docking/navigation integration as one package	Ask for AMR + battery + auto charger + docking/scheduling integration as one

7. Charging-time examples for the TIANYUE 48 V / 150 A station

The station's maximum electrical output is approximately 7.2 kW (48 V × 150 A). A simple theoretical calculation gives charging time = battery energy ÷ 7.2 kW. Actual time will be longer because charging current normally reduces near full charge and because charger/battery losses occur.

Example battery	Nominal energy	Idealized minimum time at 7.2 kW*
48 V / 100 Ah	4.8 kWh	~40 min
48 V / 200 Ah	9.6 kWh	~80 min
48 V / 300 Ah	14.4 kWh	~120 min
48 V / 400 Ah	19.2 kWh	~160 min

*These are theoretical calculations, not promised charging times. The actual charger may not deliver 150 A continuously over the whole charge cycle.

8. Recommended configuration for the project

- AMR: 600 kg preferred, low-profile, Mecanum, vertical top-platform/jacking lift.
- Battery: LiFePO4, 48 V preferred; ask supplier to size capacity for the required duty cycle rather than simply accepting an 8 h headline.

- Automatic charging: contact-docking station; automatic return to charger when battery reaches a defined threshold.
- Opportunity charging: use idle/waiting periods to reduce deep discharge and support multi-shift operation.
- Software: automatic return-to-charge, charging status monitoring, task pause/resume after charging, and fleet scheduling if multiple AMRs are used.
- Safety: charging interlock, battery protection, emergency stop, temperature monitoring and fault reporting.

9. RFQ questions to send TIANYUE and TZBOT

- What is the exact battery capacity in Ah and kWh for the 600 kg AMR?
- What is the guaranteed operating time under a defined duty cycle with 600 kg load?
- What is the recommended minimum battery state-of-charge before automatic return to charger?
- Can the AMR automatically return to the charger, dock, charge and resume the previous task?
- What is the actual charging time from 20% to 80% and 20% to 100%?
- For TIANYUE: is the AGVC-48V150A-TY charger fully compatible with the 600 kg Mecanum AMR?
- What is the EXW and FOB China price of the AMR, battery and automatic charging station separately and as a package?
- Does the quoted price include charging-station positioning, communication and software integration?
- What are warranty terms, battery cycle life, replacement battery price and spare-parts availability?
- For European import: provide CE/declaration documentation and battery transport documentation (including UN38.3/MSDS where applicable).

10. Source links

TIANYUE 48V/150A automatic charging station:
https://www.roboticsagv.com/sale-44898002-integrated-48v-150a-agv-battery-charger-automatic-retractable-charging-pile.html

TIANYUE 300 kg / 8 h battery benchmark: https://www.roboticsagv.com/doc/54447656/navigation-accuracy-5-10mm-agv-automated-guided-vehicle-with-machine-size-792-x-580-x-250-mm-and-com.pdf

TIANYUE 8 h automatic-return-to-charge AGV:
https://www.roboticsagv.com/doc/54244643/agv-automated-guided-vehicle-with-100kg-1000kg-rated-load-8-hours-battery-life-and-stainless-steel-s.pdf

TIANYUE 8 h back-lifting AGV:
https://www.roboticsagv.com/doc/54419335/8-hours-batterylife-agv-automated-guided-vehicle-with-differential-drive-and-550mm-rotation-radius-f.pdf

TZBOT 600 kg Laser-SLAM AMR: https://www.tzbotautomation.com/automatic-guided-vehicle/lifting-agv/tzbot-lidar-guided-autonomous-intelligent-amr.html

TZBOT automatic charging solution: https://www.tzbotautomation.com/info/tzbot-provide-intelligent-handling-solutions-55029911.html

TZBOT 600 kg magnetic AGV with auto charge:
https://www.tzbotautomation.com/automatic-guided-vehicle/lifting-agv/600kg-carrying-capacity-magnetic-agv-robot.html

TZBOT laser-sensor AMR battery benchmark: https://www.tzbotautomation.com/automatic-guided-vehicle/laser-agv/tzbot-laser-sensor-amr-robot.html

Note: marketplace prices cited in this report are benchmarks found during research; exact supplier quotations can change and should be confirmed directly before purchase.